Lab 1: Packet analysis at application layer using Wireshark

SCSR1213 Network Communications

Universiti Teknologi Malaysia

Objective:

1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.

2. To introduce student with Wireshark software tool for packet analyzer.

3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

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PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures ("sniffs") messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

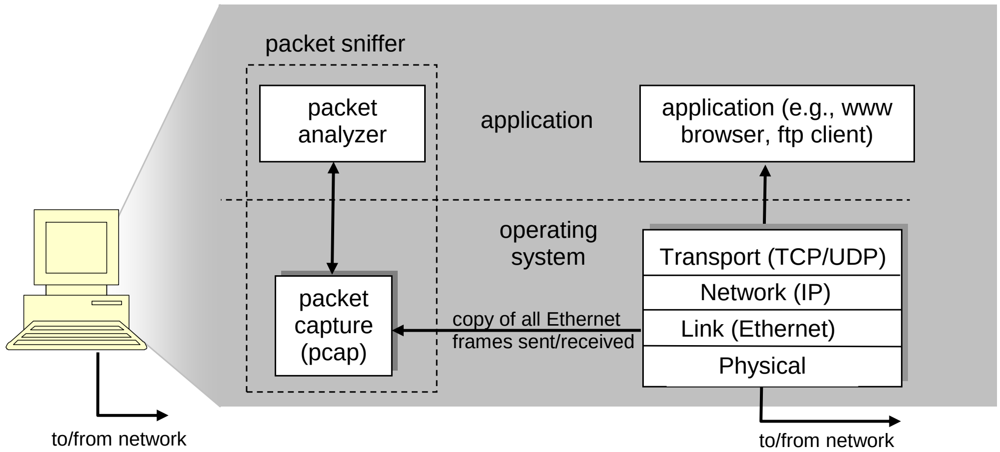


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must "understand" the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string "GET," "POST," or "HEAD".

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

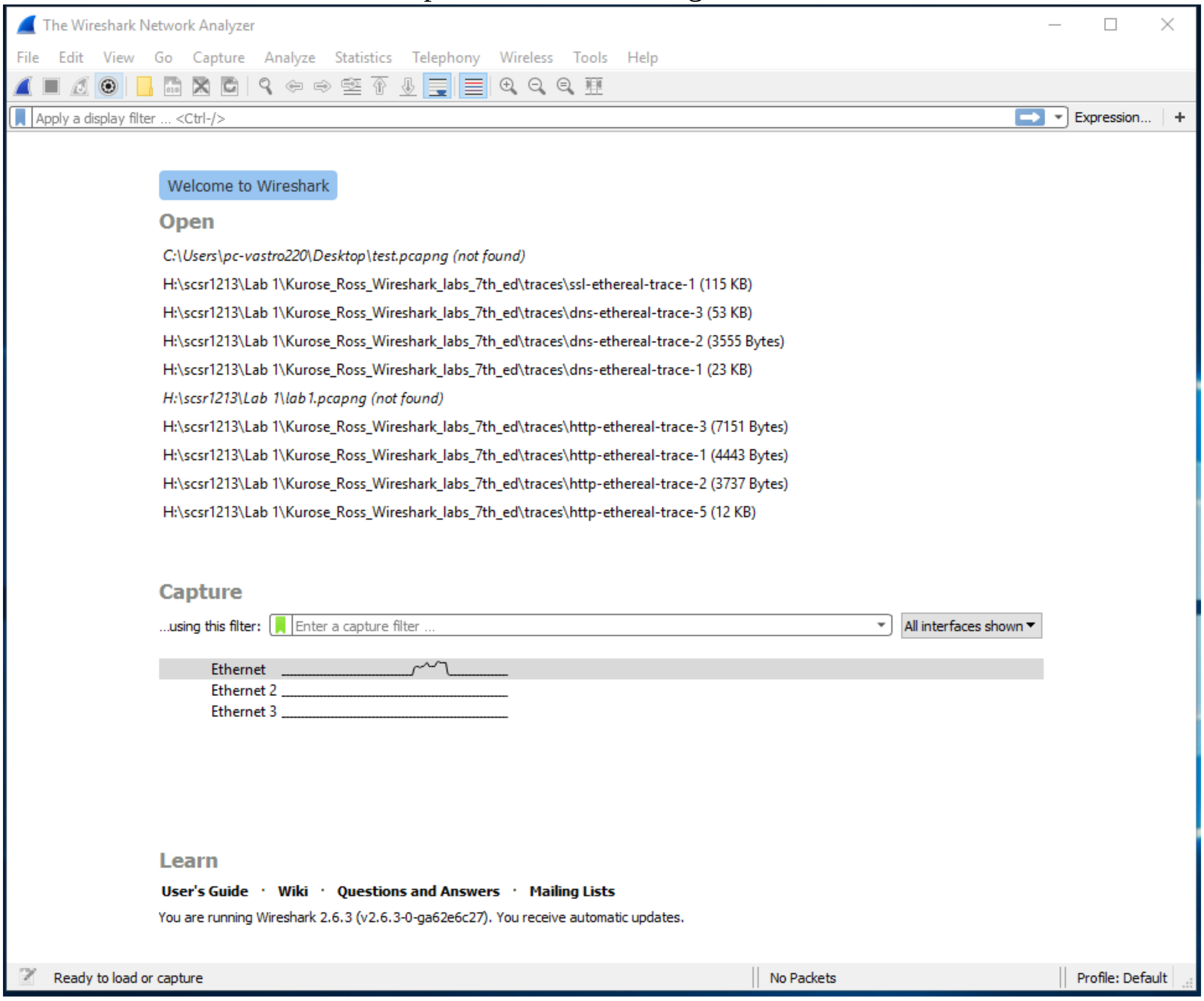


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

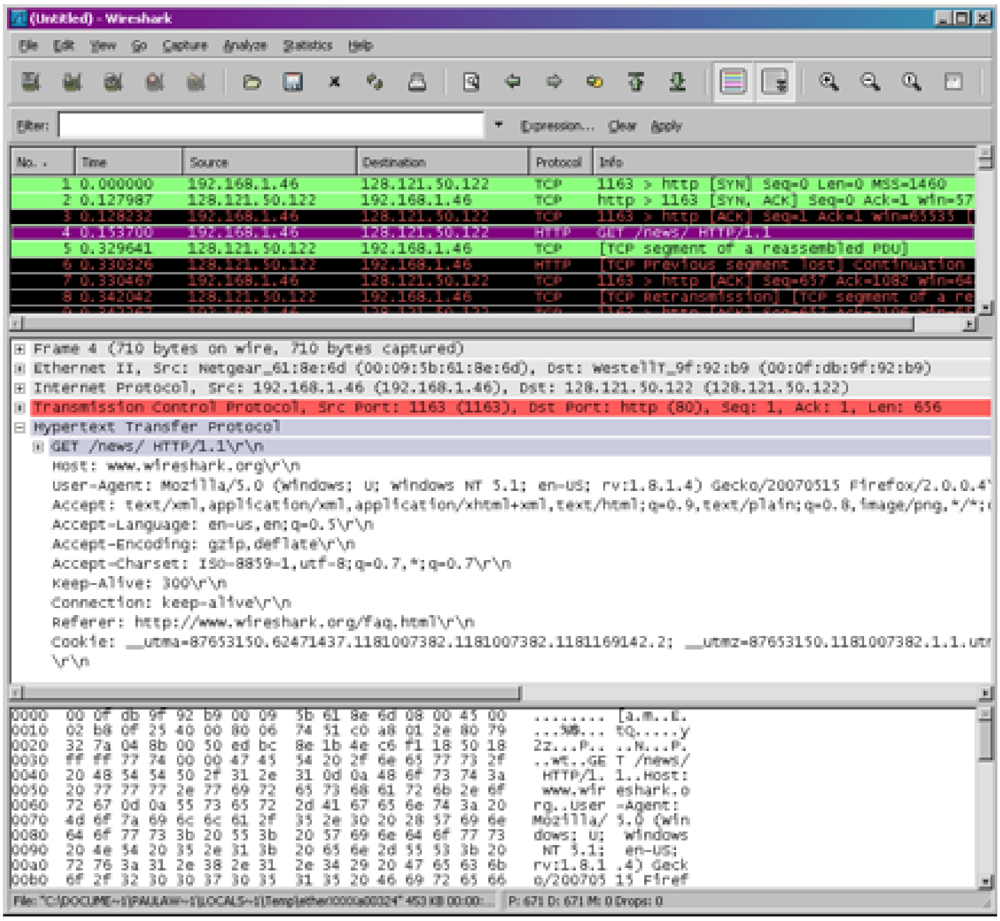


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field,** into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

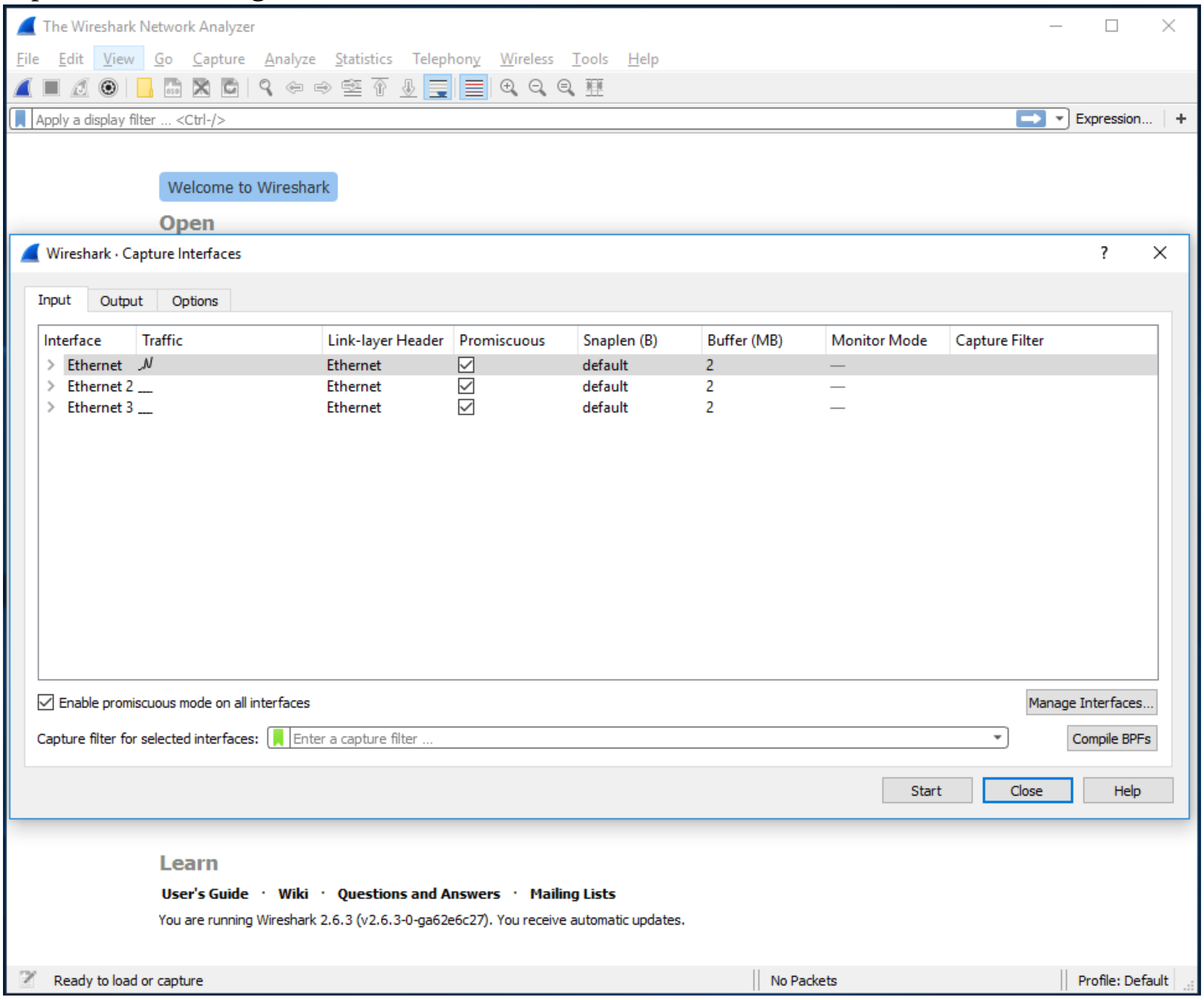


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.

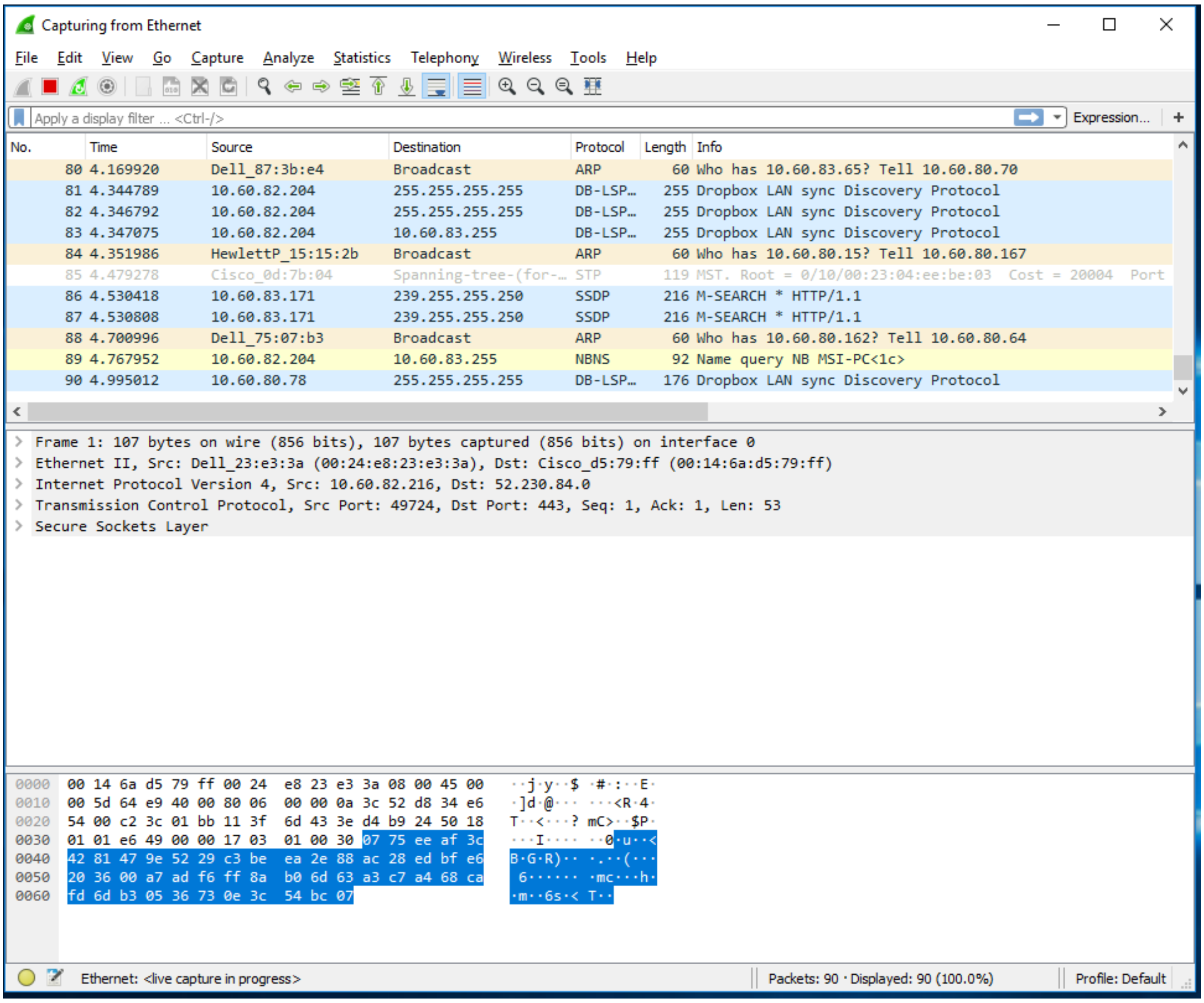


Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

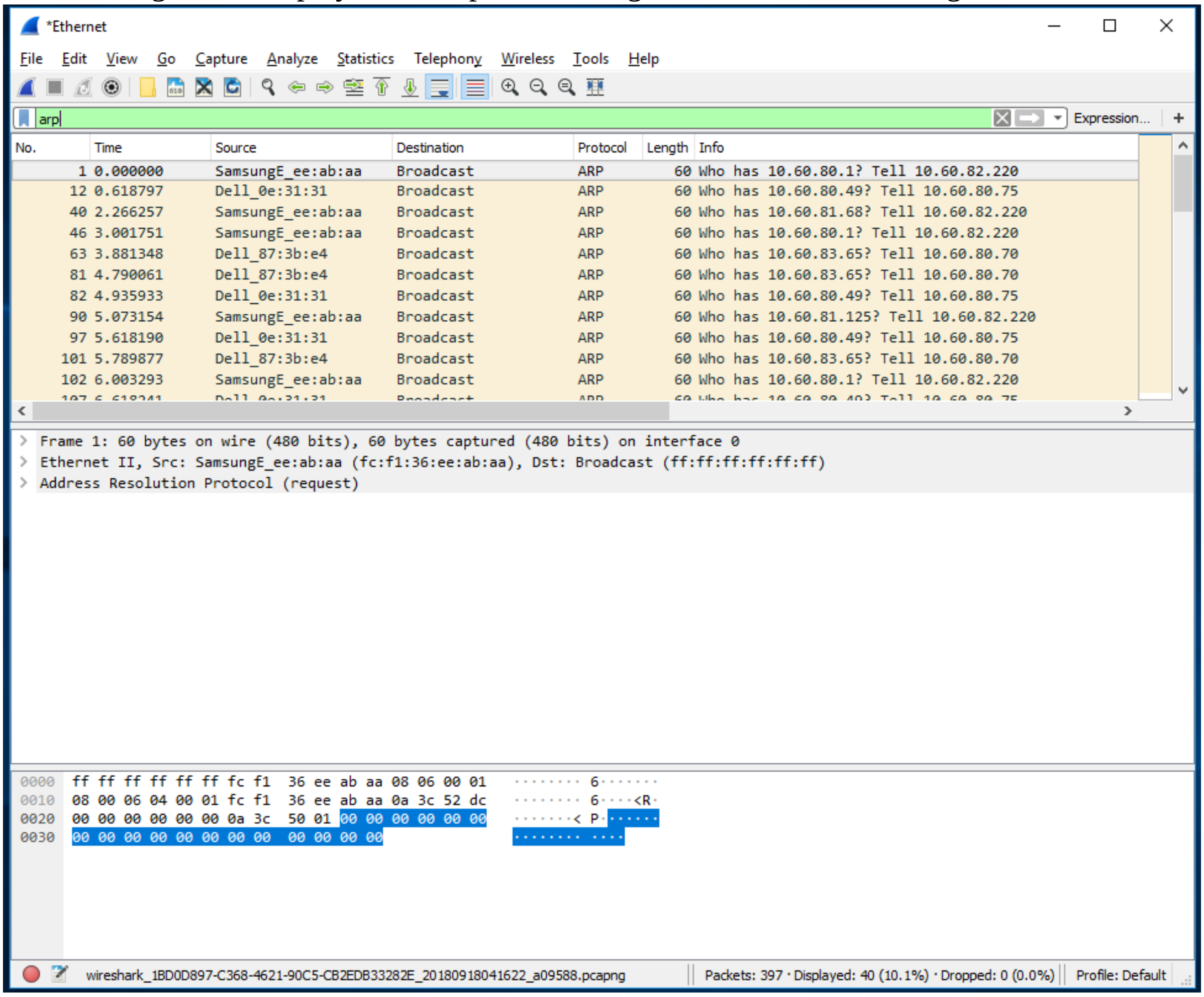


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

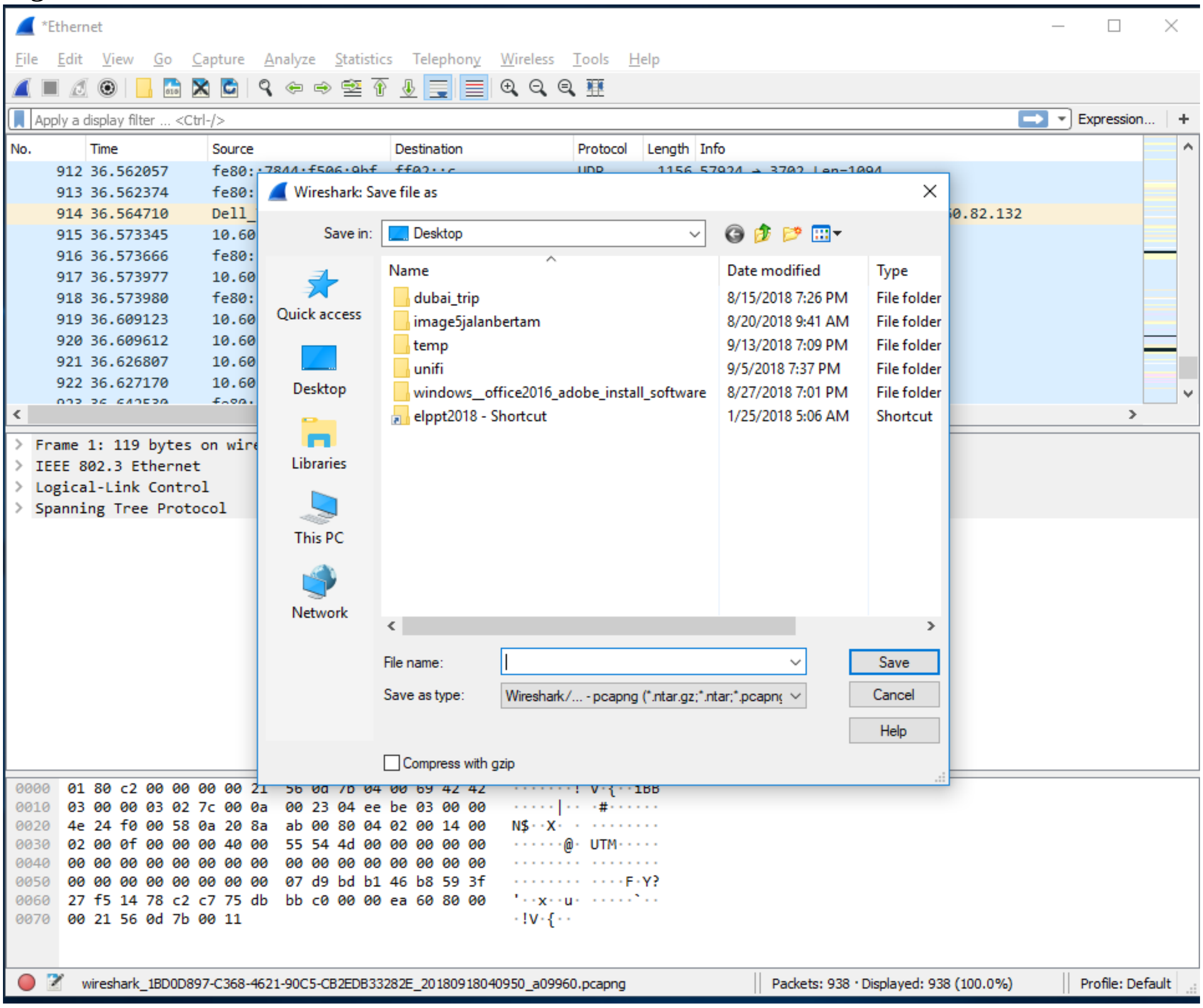


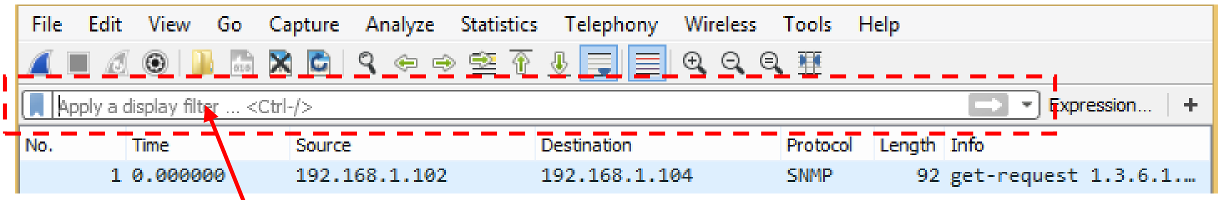
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter "**http**" (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



packet display filter

- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. What version of HTTP is the server running?
   HTTP Version 1

No.	Time	Source	Destination	Protocol	Length	Info
10	4.694850	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-1.html HTTP/1.1
12	4.718993	128.119.245.12	192.168.1.102	HTTP	439	HTTP/1.1 200 OK (text/html)
13	4.724332	192.168.1.102	128.119.245.12	HTTP	541	GET /favicon.ico HTTP/1.1
14	4.750366	128.119.245.12	192.168.1.102	HTTP	1395	HTTP/1.1 404 Not Found (text/html)

2. What is the IP address of the client computer?
   192.168.1.102

No.	Time	Source	Destination	Protocol	Length	Info
10	4.694850	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-1.html HTTP/1.1
12	4.718993	128.119.245.12	192.168.1.102	HTTP	439	HTTP/1.1 200 OK (text/html)
13	4.724332	192.168.1.102	128.119.245.12	HTTP	541	GET /favicon.ico HTTP/1.1
14	4.750366	128.119.245.12	192.168.1.102	HTTP	1395	HTTP/1.1 404 Not Found (text/html)

3. What is the IP address of the gaia.cs.umass.edu server?
   128.119.245.12

No.	Time	Source	Destination	Protocol	Length	Info
10	4.694850	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-1.html HTTP/1.1
12	4.718993	128.119.245.12	192.168.1.102	HTTP	439	HTTP/1.1 200 OK (text/html)
13	4.724332	192.168.1.102	128.119.245.12	HTTP	541	GET /favicon.ico HTTP/1.1
14	4.750366	128.119.245.12	192.168.1.102	HTTP	1395	HTTP/1.1 404 Not Found (text/html)

4. How many bytes of content are being returned to client browser?
   73 bytes
   1071 bytes

```
Hypertext Transfer Protocol
  HTTP/1.1 200 OK\r\n
  Date: Tue, 23 Sep 2003 05:29:50 GMT\r\n
  Server: Apache/2.0.40 (Red Hat Linux)\r\n
  Last-Modified: Tue, 23 Sep 2003 05:29:00 GMT\r\n
  ETag: "1bfed-49-79d5bf00"\r\n
  Accept-Ranges: bytes\r\n
  Content-Length: 73\r\n
  Keep-Alive: timeout=10, max=100\r\n
  Connection: Keep-Alive\r\n
  Content-Type: text/html; charset=ISO-8859-1\r\n
  \r\n
  [Request in frame: 10]
  [Time since request: 0.024143000 seconds]
  [Request URI: /ethereal-labs/lab2-1.html]
  [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-1.html]
  File Data: 73 bytes
```

```
Hypertext Transfer Protocol
  HTTP/1.1 404 Not Found\r\n
  Date: Tue, 23 Sep 2003 05:29:50 GMT\r\n
  Server: Apache/2.0.40 (Red Hat Linux)\r\n
  Vary: accept-language\r\n
  Accept-Ranges: bytes\r\n
  Content-Length: 1071\r\n
  Keep-Alive: timeout=10, max=99\r\n
  Connection: Keep-Alive\r\n
  Content-Type: text/html; charset=ISO-8859-1\r\n
  \r\n
  [Request in frame: 13]
  [Time since request: 0.026034000 seconds]
  [Request URI: /favicon.ico]
  [Full request URI: http://gaia.cs.umass.edu/favicon.ico]
  File Data: 1071 bytes
```

5. What is the status code returned from the server to client browser?
   200(OK) and 404 (Not Found)

No.	Time	Source	Destination	Protocol	Length	Info
10	4.694850	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-1.html HTTP/1.1
12	4.718993	128.119.245.12	192.168.1.102	HTTP	439	HTTP/1.1 200 OK (text/html)
13	4.724332	192.168.1.102	128.119.245.12	HTTP	541	GET /favicon.ico HTTP/1.1
14	4.750366	128.119.245.12	192.168.1.102	HTTP	1395	HTTP/1.1 404 Not Found (text/html)

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?

```
> Frame 8: 555 bytes on wire (4440 bits), 555 bytes captured (4440 bits) on interface unknown, id 0
> Ethernet II, Src: Dell_4f:36:23 (00:08:74:4f:36:23), Dst: LinksysGroup_da:af:73 (00:06:25:da:af:73)
> Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 4247, Dst Port: 80, Seq: 1, Ack: 1, Len: 501
v Hypertext Transfer Protocol
  > GET /ethereal-labs/lab2-2.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r\
    Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,i
    Accept-Language: en-us, en;q=0.50\r\n
    Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
    Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66\r\n
    Keep-Alive: 300\r\n
    Connection: keep-alive\r\n
    \r\n
    [Response in frame: 10]
    [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-2.html]
```

http

No.	Time	Source	Destination	Protocol	Length	Info
8	2.331268	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-2.html HTTP/1.1
10	2.357902	128.119.245.12	192.168.1.102	HTTP	739	HTTP/1.1 200 OK (text/html)
14	5.517390	192.168.1.102	128.119.245.12	HTTP	668	GET /ethereal-labs/lab2-2.html HTTP/1.1
15	5.540216	128.119.245.12	192.168.1.102	HTTP	243	HTTP/1.1 304 Not Modified

2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?

No.	Time	Source	Destination	Protocol	Length	Info
8	2.331268	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-2.html HTTP/1.1
10	2.357902	128.119.245.12	192.168.1.102	HTTP	739	HTTP/1.1 200 OK (text/html)
14	5.517390	192.168.1.102	128.119.245.12	HTTP	668	GET /ethereal-labs/lab2-2.html HTTP/1.1
15	5.540216	128.119.245.12	192.168.1.102	HTTP	243	HTTP/1.1 304 Not Modified

Yes, the server returns the content of the file as the status code shown 200 OK. The server returns the contents of the file as the content returned to my browser is the same as the content in the server.

```
Line-based text data: text/html (10 lines)
  \n
  <html>\n
  \n
  Congratulations again!  Now you've downloaded the file lab2-2.html. <br>\n
  This file's last modification date will not change.  <p>\n
  Thus  if you download this multiple times on your browser, a complete copy <br>\n
  will only be sent once by the server due to the inclusion of the IN-MODIFIED-SINCE<br>\n
  field in your browser's HTTP GET request to the server.\n
  \n
  </html>\n
```

3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?
   YES. The information follows the “IF-MODIFIED-SINCE: “header is “Tue,23 Sep 2003 05:35:00”

```
> Frame 14: 668 bytes on wire (5344 bits), 668 bytes captured (5344 bits) on interface unknown, id 0
> Ethernet II, Src: Dell_4f:36:23 (00:08:74:4f:36:23), Dst: LinksysGroup_da:af:73 (00:06:25:da:af:73)
> Internet Protocol Version 4, Src: 192.168.1.102, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 4247, Dst Port: 80, Seq: 502, Ack: 686, Len: 614
v Hypertext Transfer Protocol
  > GET /ethereal-labs/lab2-2.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.0.2) Gecko/20021120 Netscape/7.01\r
    Accept: text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,video/x-mng,
    Accept-Language: en-us, en;q=0.50\r\n
    Accept-Encoding: gzip, deflate, compress;q=0.9\r\n
    Accept-Charset: ISO-8859-1, utf-8;q=0.66, *;q=0.66\r\n
    Keep-Alive: 300\r\n
    Connection: keep-alive\r\n
    If-Modified-Since: Tue, 23 Sep 2003 05:35:00 GMT\r\n
    If-None-Match: "1bfef-173-8f4ae900"\r\n
    Cache-Control: max-age=0\r\n
    \r\n
    [Response in frame: 15]
    [Full request URI: http://gaia.cs.umass.edu/ethereal-labs/lab2-2.html]
```

4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

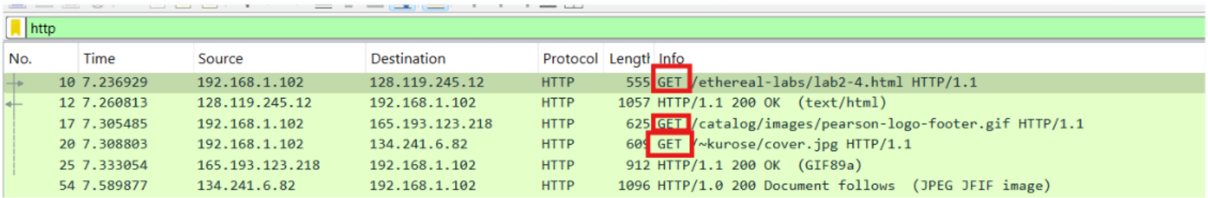
   304 Not Modified. The server did not return the content of the file because it shows Not Modified which means the server did not send the updated content of the file, it retrieved the previous version from the browser’s cache memory.

No.	Time	Source	Destination	Protocol	Length	Info
8	2.331268	192.168.1.102	128.119.245.12	HTTP	555	GET /ethereal-labs/lab2-2.html HTTP/1.1
10	2.357902	128.119.245.12	192.168.1.102	HTTP	739	HTTP/1.1 200 OK (text/html)
14	5.517390	192.168.1.102	128.119.245.12	HTTP	668	GET /ethereal-labs/lab2-2.html HTTP/1.1
15	5.540216	128.119.245.12	192.168.1.102	HTTP	243	HTTP/1.1 304 Not Modified

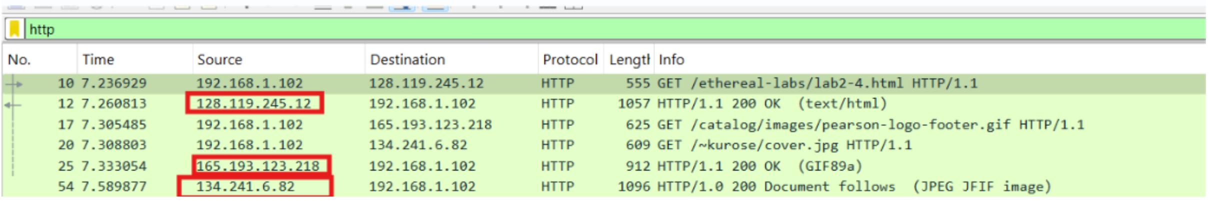
B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

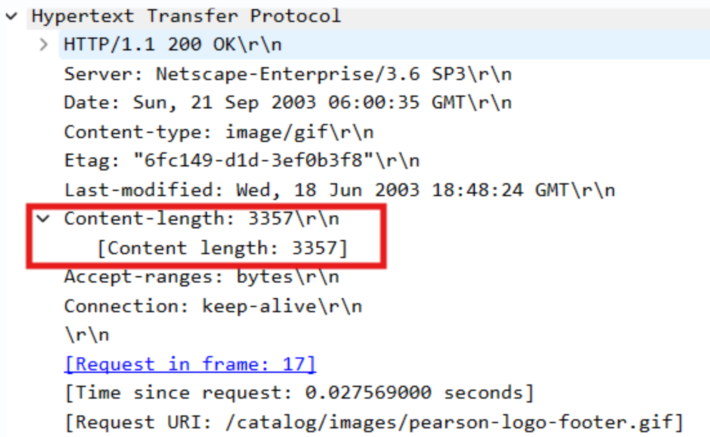
1. How many HTTP GET request messages did client browser send?
   3 HTTP GET messages



2. To which Internet addresses were these GET requests sent?
   128.119.245.12
   165.193.123.218
   134.241.6.82



3. How many bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?



4. How many bytes of content are being returned to the client browser for the **cover.jpg** image file?

```
v Hypertext Transfer Protocol
  > HTTP/1.0 200 Document follows\r\n
    Date: Tue, 23 Sep 2003 05:38:44 GMT\r\n
    Server: NCSA/1.5.2\r\n
    Last-modified: Tue, 23 Sep 2003 04:56:38 GMT\r\n
    Content-type: image/jpeg\r\n
  v Content-length: 15642\r\n
      [Content length: 15642]
    \r\n
    [Request in frame: 20]
    [Time since request: 0.281074000 seconds]
    [Request URI: /~kurose/cover.jpg]
    [Full request URI: http://manic.cs.umass.edu/~kurose/cover.jpg]
    File Data: 15642 bytes
```

PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2

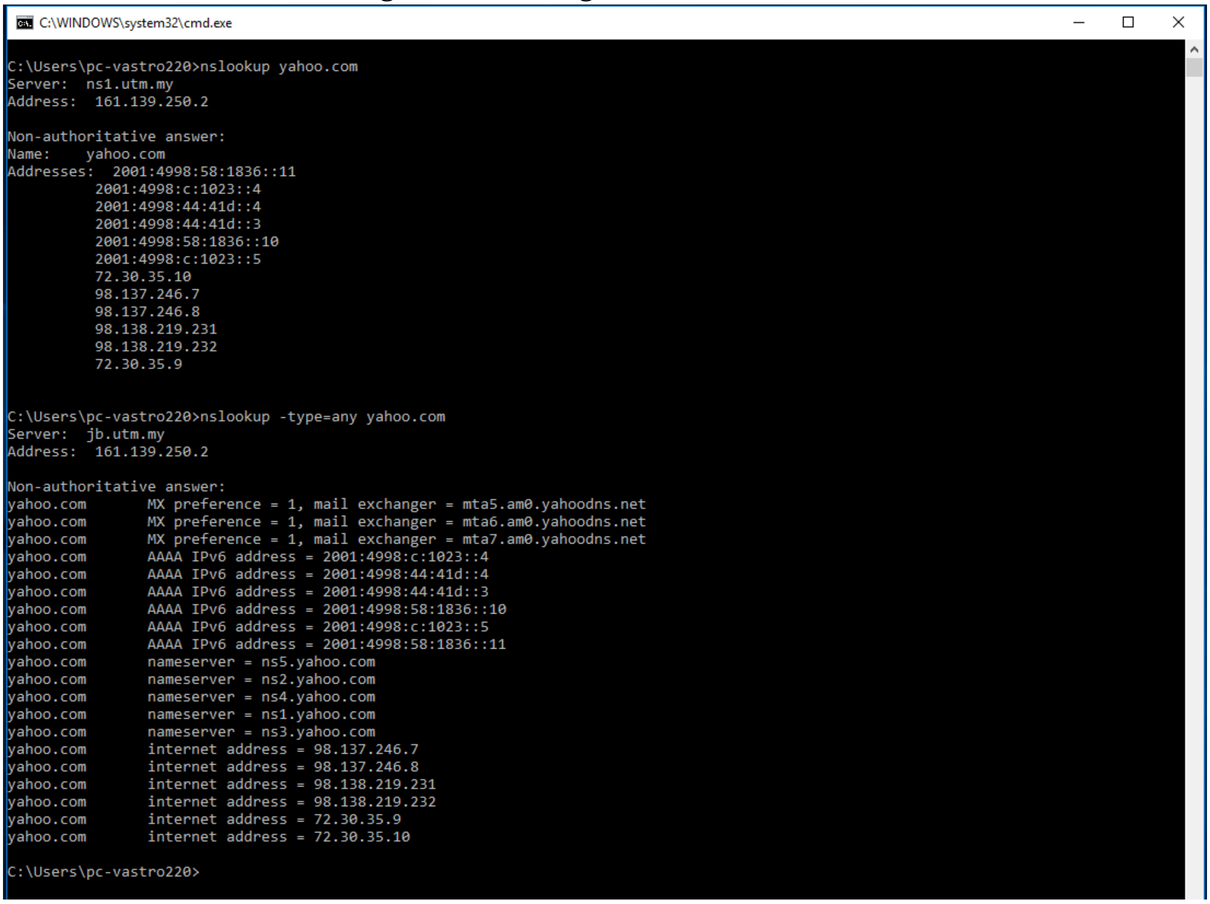


Figure C.1: nslookup result

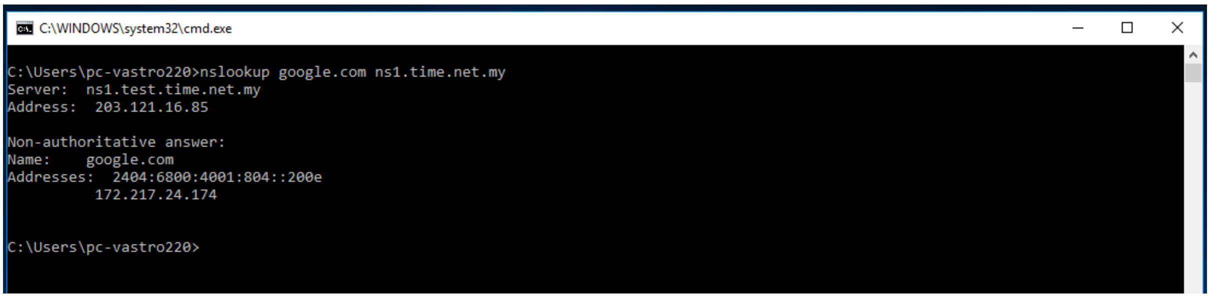
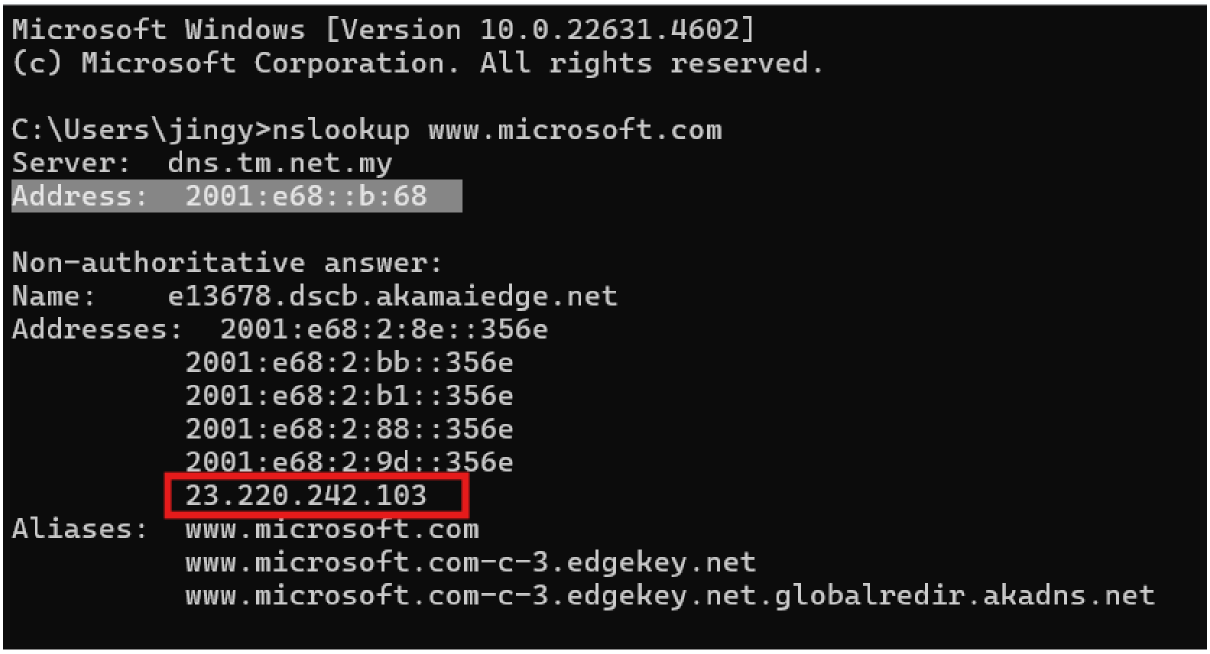


Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

   23.220.242.103



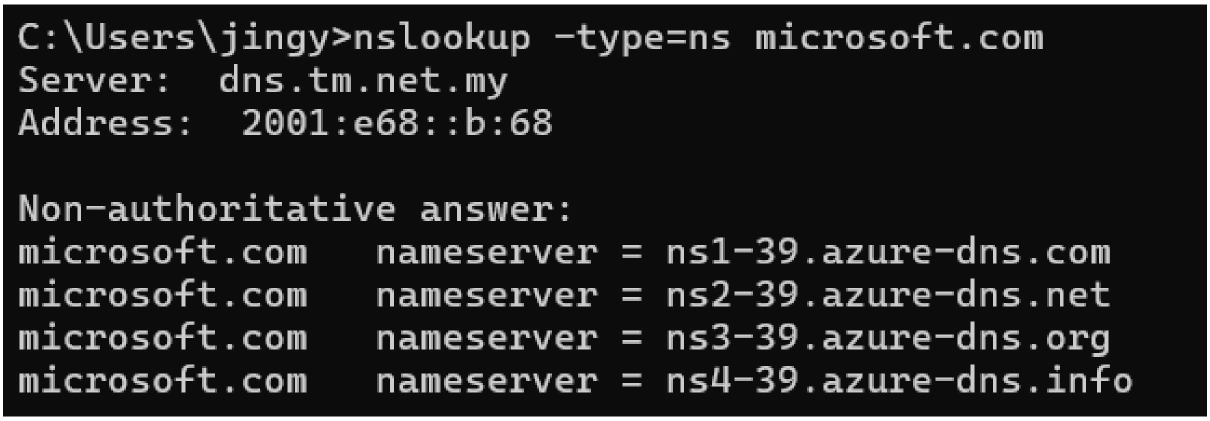
```
Microsoft Windows [Version 10.0.22631.4602]
(c) Microsoft Corporation. All rights reserved.

C:\Users\jingy>nslookup www.microsoft.com
Server:  dns.tm.net.my
Address:  2001:e68::b:68

Non-authoritative answer:
Name:    e13678.dscb.akamaiedge.net
Addresses:  2001:e68:2:8e::356e
          2001:e68:2:bb::356e
          2001:e68:2:b1::356e
          2001:e68:2:88::356e
          2001:e68:2:9d::356e
          23.220.242.103
Aliases:  www.microsoft.com
          www.microsoft.com-c-3.edgekey.net
          www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshots to your answer.

   ns1-39.azure-dns.com
   ns2-39.azure-dns.net
   ns3-39.azure-dns.org
   ns4-39.azure-dns.info



```
C:\Users\jingy>nslookup -type=ns microsoft.com
Server:  dns.tm.net.my
Address:  2001:e68::b:68

Non-authoritative answer:
microsoft.com   nameserver = ns1-39.azure-dns.com
microsoft.com   nameserver = ns2-39.azure-dns.net
microsoft.com   nameserver = ns3-39.azure-dns.org
microsoft.com   nameserver = ns4-39.azure-dns.info
```

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

- Information about host, use the following command: ipconfig /all

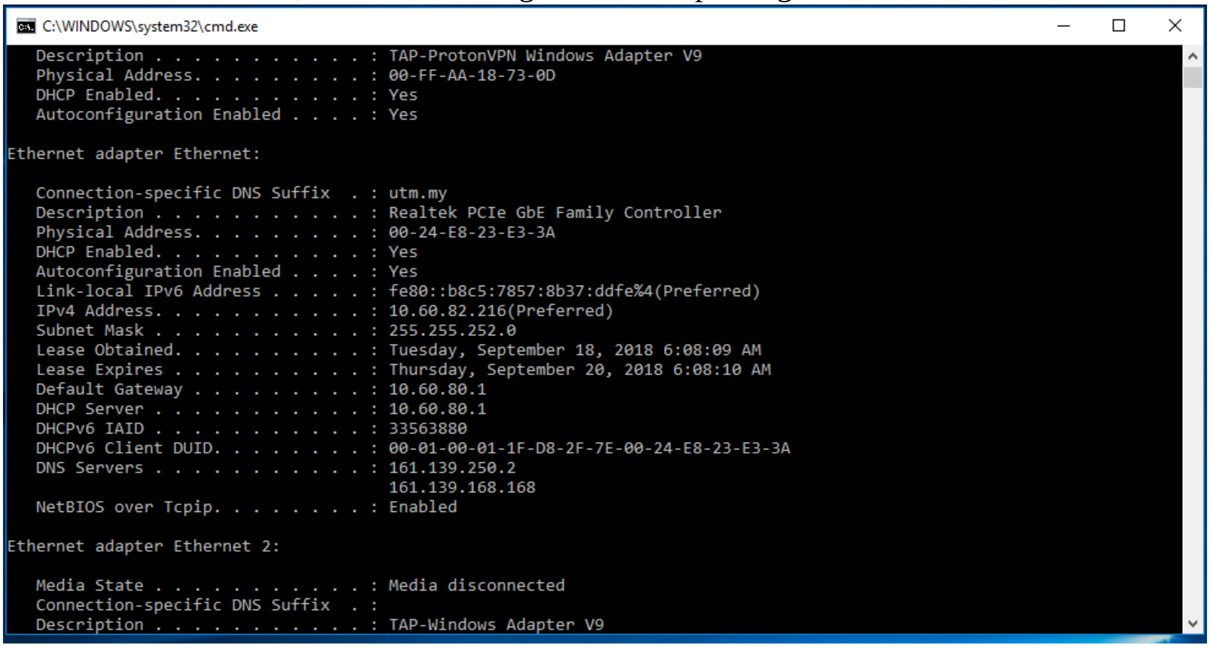


Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds.
  Command: ipconfig /displaydns

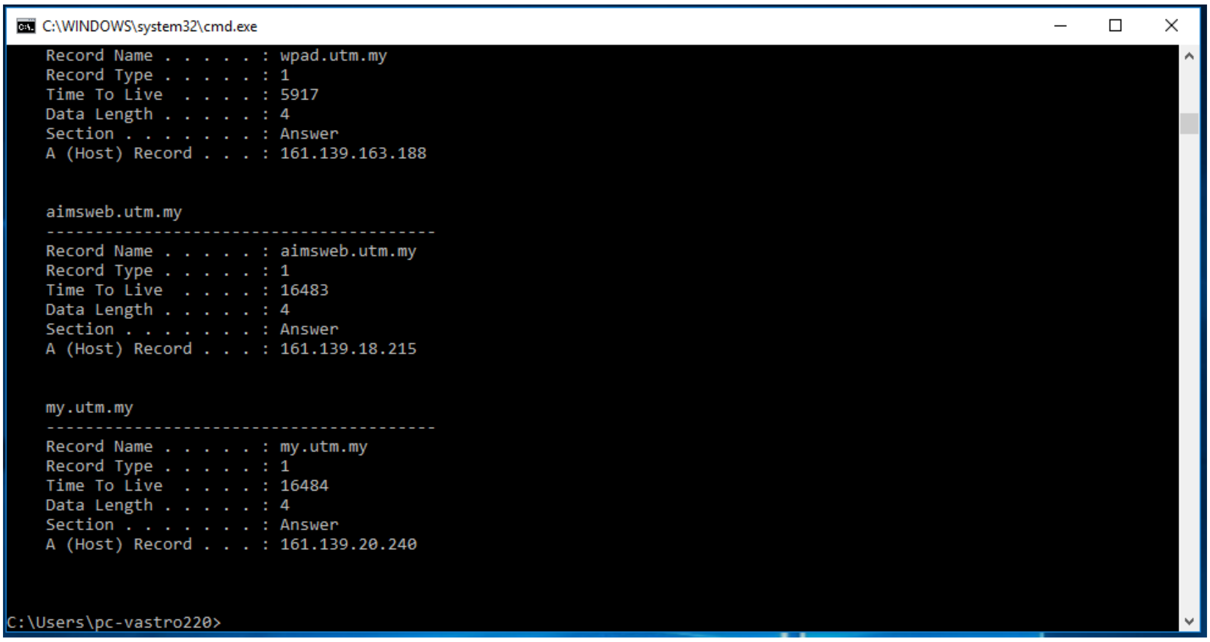


Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

Command: ipconfig /flushdns

```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

C:\Users\pc-vastro220>
```

Figure C.5: ipconfig /flushdns result

3.0 Tracing DNS with Wireshark

- Open packet trace file dns-trace-1. Answer the following questions.

1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.
   Query message: Frame 8
   Response Message: Frame 9
   They are sent over UDP

```
dns
No.  Time      Source          Destination     Protocol Length Info
8    3.075845  128.238.38.160  128.238.29.23   DNS      72  Standard query 0x006e A www.ietf.org
9    3.076689  128.238.29.23   128.238.38.160  DNS      104 Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51
```

```
> Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
> Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
> User Datagram Protocol, Src Port: 3163, Dst Port: 53
> Domain Name System (query)
```

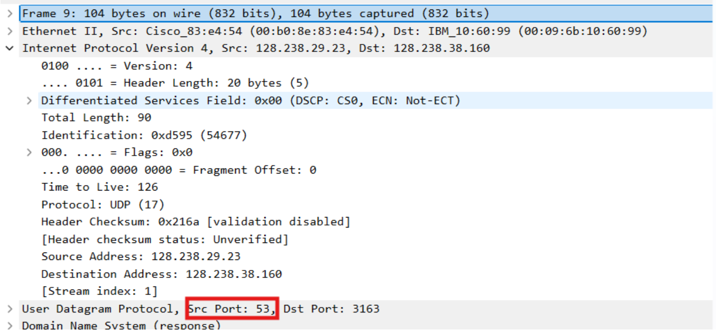
```
> Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits)
> Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
> Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160
> User Datagram Protocol, Src Port: 53, Dst Port: 3163
> Domain Name System (response)
```

2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

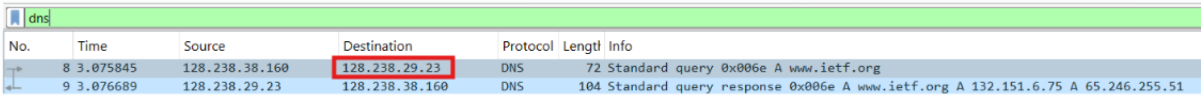
   The destination port for the DNS query message is port 53

```
> Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
v Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 58
    Identification: 0x229e (8862)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0xd281 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 128.238.38.160
    Destination Address: 128.238.29.23
    [Stream index: 1]
> User Datagram Protocol, Src Port: 3163, Dst Port: 53
> Domain Name System (query)
```

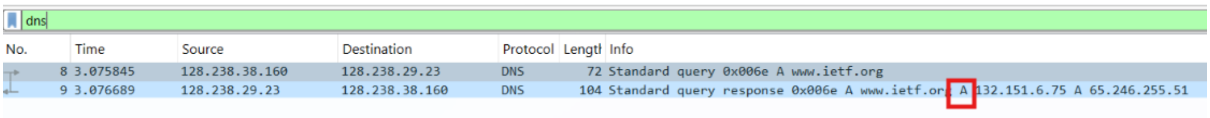
The source port for the DNS response message is port 53

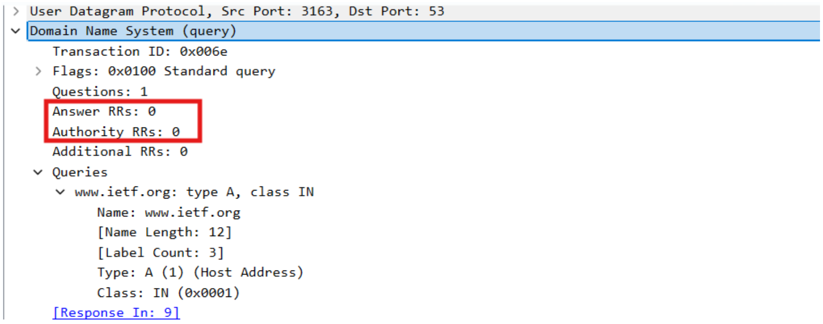


3. To what IP address is the DNS query message sent? Add screenshots in your answer.
   128.238.29.23



4. Examine the DNS query message. What "Type" of DNS query is it? Does the query message contain any "answers"? Add screenshots in your answer.
   Type A. The query doesn't contain any answer.





5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

   2 Answer. Each of the answers contain domain name, type of address, class, IP address, time to live and data length.

```
> User Datagram Protocol, Src Port: 53, Dst Port: 3163
v Domain Name System (response)
    Transaction ID: 0x006e
  > Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 2
    Authority RRs: 0
    Additional RRs: 0
  v Queries
    v www.ietf.org: type A, class IN
        Name: www.ietf.org
        [Name Length: 12]
        [Label Count: 3]
        Type: A (1) (Host Address)
        Class: IN (0x0001)
  v Answers
    > www.ietf.org: type A, class IN, addr 132.151.6.75
    > www.ietf.org: type A, class IN, addr 65.246.255.51
    [Request In: 8]
```

```
v Answers
  v www.ietf.org: type A, class IN, addr 132.151.6.75
      Name: www.ietf.org
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 1678 (27 minutes, 58 seconds)
      Data length: 4
      Address: 132.151.6.75
  v www.ietf.org: type A, class IN, addr 65.246.255.51
      Name: www.ietf.org
      Type: A (1) (Host Address)
      Class: IN (0x0001)
      Time to live: 1678 (27 minutes, 58 seconds)
      Data length: 4
      Address: 65.246.255.51
```

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.
   The destination IP address of the SYN packet (132.151.6.75) corresponds to the IP provided in the DNS response message

tcp

No.	Time	Source	Destination	Protocol	Length	Info
10	3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
11	3.096413	132.151.6.75	128.238.38.160	TCP	62	80 → 3369 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
12	3.096463	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429	GET / HTTP/1.1
14	3.111678	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=0
15	3.120640	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
16	3.128093	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1381 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
17	3.128148	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=2761 Win=64860 Len=0
18	3.148016	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=2761 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
19	3.148069	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=4141 Win=64860 Len=0
20	3.153211	132.151.6.75	128.238.38.160	HTTP	1055	HTTP/1.1 200 OK (text/html)
21	3.153293	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=5143 Win=63859 Len=0
22	3.161867	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [FIN, ACK] Seq=376 Ack=5143 Win=63859 Len=0
23	3.174716	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=5143 Ack=377 Win=6432 Len=0
24	3.178159	128.238.38.160	132.151.6.75	TCP	62	3370 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM

```
> Frame 10: 62 bytes on wire (496 bits), 62 bytes captured (496 bits)
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
> Internet Protocol Version 4, Src: 128.238.38.160, Dst: 132.151.6.75
v Transmission Control Protocol, Src Port: 3369, Dst Port: 80, Seq: 0, Len: 0
```

7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?

No,it is because the DNS query will only appear once in the entire tracing. The host did not initiate a new DNS query between lines 10 and 27.

10	3.078479	128.238.38.160	132.151.6.75	TCP	62	3369 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
11	3.096413	132.151.6.75	128.238.38.160	TCP	62	80 → 3369 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
12	3.096463	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
13	3.096708	128.238.38.160	132.151.6.75	HTTP	429	GET / HTTP/1.1
14	3.111678	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=0
15	3.120640	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
16	3.128093	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=1381 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
17	3.128148	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=2761 Win=64860 Len=0
18	3.148016	132.151.6.75	128.238.38.160	TCP	1434	80 → 3369 [ACK] Seq=2761 Ack=376 Win=6432 Len=1380 [TCP PDU reassembled in 20]
19	3.148069	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=4141 Win=64860 Len=0
20	3.153211	132.151.6.75	128.238.38.160	HTTP	1055	HTTP/1.1 200 OK (text/html)
21	3.153293	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [ACK] Seq=376 Ack=5143 Win=63859 Len=0
22	3.161867	128.238.38.160	132.151.6.75	TCP	54	3369 → 80 [FIN, ACK] Seq=376 Ack=5143 Win=63859 Len=0
23	3.174716	132.151.6.75	128.238.38.160	TCP	60	80 → 3369 [ACK] Seq=5143 Ack=377 Win=6432 Len=0
24	3.178159	128.238.38.160	132.151.6.75	TCP	62	3370 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
25	3.179283	128.238.38.160	132.151.6.75	TCP	62	3371 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM
26	3.191649	132.151.6.75	128.238.38.160	TCP	62	80 → 3370 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
27	3.191726	128.238.38.160	132.151.6.75	TCP	54	3370 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0
28	3.191998	128.238.38.160	132.151.6.75	HTTP	320	GET /images/ietflogo2e.gif HTTP/1.1
29	3.192665	132.151.6.75	128.238.38.160	TCP	62	80 → 3371 [SYN, ACK] Seq=0 Ack=1 Win=5840 Len=0 MSS=1380 SACK_PERM
30	3.192695	128.238.38.160	132.151.6.75	TCP	54	3371 → 80 [ACK] Seq=1 Ack=1 Win=64860 Len=0

- Open packet trace file dns-trace-2 for nslookup.
- We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
- Answer the following questions.

8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.
   The destination port for query message is 53.

No.	Time	Source	Destination	Protocol	Lengtl	Info
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR dns-prime.poly.edu
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu SOA dns-prime.poly.edu
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS STRAWB.mit.edu NS W20NS.mit.edu A 18.72.0.3 A 18.71.0.1…

```
> Frame 19: 71 bytes on wire (568 bits), 71 bytes captured (568 bits)
> Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
> Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.22
> User Datagram Protocol, Src Port: 3742, Dst Port: 53
v Domain Name System (query)
    Transaction ID: 0x0003
  > Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  > Queries
    [Response In: 20]
```

9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.

   DNS query message sent to 128.238.29.22 IP address.The IP address is not my default local DNS server.

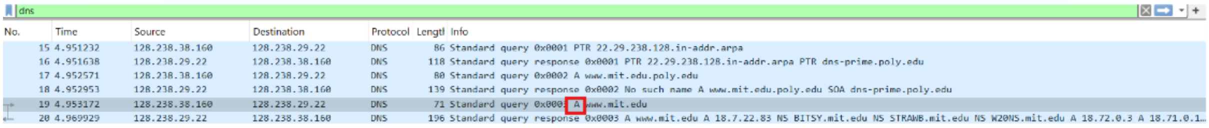
dns

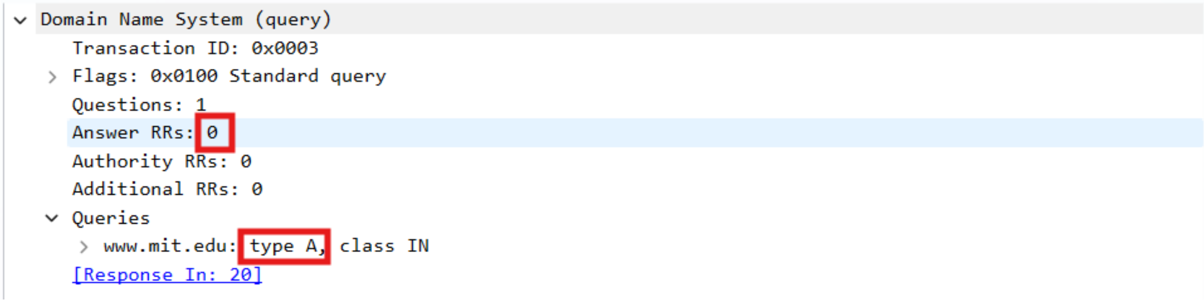
No.	Time	Source	Destination	Protocol	Lengtl	Info
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR dns-prime.poly.edu
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu SOA dns-prime.poly.edu
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS STRAWB.mit.edu NS W20NS.mit.edu A 18.72.0.3 A 18.71.0.1…

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Killer(R) Wi-Fi 6E AX1675i 160MHz Wireless Network Adapter (211NGW)
   Physical Address. . . . . . . . . : 10-91-D1-ED-DD-D9
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2001:e68:5407:54a7:a582:a672:70bd:f321(Preferred)
   Temporary IPv6 Address. . . . . . : 2001:e68:5407:54a7:7085:b5c9:b1b0:52bb(Preferred)
   Link-local IPv6 Address . . . . . : fe80::752c:d292:c0c9:2ee%12(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.0.182(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Sunday, 19 January, 2025 11:43:48 AM
   Lease Expires . . . . . . . . . . : Monday, 20 January, 2025 1:00:39 PM
   Default Gateway . . . . . . . . . : fe80::1%12
                                       192.168.0.1
   DHCP Server . . . . . . . . . . . : 192.168.0.1
   DHCPv6 IAID . . . . . . . . . . . : 118526417
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2E-2B-48-1A-38-A7-46-02-F1-8B
   DNS Servers . . . . . . . . . . . : fe80::1%12
                                       192.168.0.1
   Primary WINS Server . . . . . . . : 192.168.0.1
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

10. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.
    Type A. No answer





11. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.
    1 answer which contain domain name, type of address, the class, time to live, data length and IP address